

# Classification of Four Categories of EEG Signals Based on Relevance Vector Machine

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**Abstract**—EEG classification is an important signal acquisition equipment of brain-interface research and application. In this paper, an improved relevance vector machine (RVM) is proposed to classify four-class motor imagery EEG signals. The original EEG signals are processed by 3-24Hz band-pass filter. Thereafter, EEG feature vectors are extracted from the band-pass filtered EEG signals with one versus one common spatial patterns (OVO-CSP). Then, the improved RVM algorithm with kernel function, which combines Gaussian kernel function and Cauchy kernel function, is applied to classify the EEG signals. A public dataset (BCI Competition IV-II-a) is employed to verify the proposed improved kernel function method. Five-fold cross-validation was used to ensure that the classification of the experiment is more credible. The classification results of the combined kernel function are compared with results of the Gaussian kernel function and the Cauchy kernel function. The experimental results show that the highest classification accuracy of the proposed kernel function and the single kernel function in the first data set are 64.40% and 60.60%, respectively, and in the second data sets are 67.58% and 63.33%. The average classification accuracy of the proposed kernel function is 4%-4.19% higher than that of the single kernel function. These results show that the proposed method is advantageous in the classification of four-class motor imagery.

**Index Terms** – RVM. OVO-CSP. Motor imagery. Single kernel function. Combined kernel function.

## I. INTRODUCTION

Brain-Computer Interface (BCI) is a technology that uses external auxiliary equipment to establish communication and control channel between human brain and computer or other electronic equipment [1-2]. A typical BCI system generally consists of signal acquisition, signal processing and controller. What study in this paper belongs to the second one which is vital important for research and application of BCI. Electroencephalogram (EEG) is one of popular means used by researchers to study brain activity. The researchers addressing the original EEG signal to remove the artifacts, and then through the feature extraction and classification algorithm to identify the subject's "idea" hidden in the EEG signals [3-4].

Researchers aim to obtain the implicit functional relationship from the training data by classification algorithm, and use the function to yield the mapping result of the new test data. The typical two classifier includes linear discriminant

analysis (LDA), support vector machine (SVM), artificial neural network (ANN) and so on. LDA using the method of dimensionality reduction to distinguish two-class signals. In reference [5], the author present an unsupervised adaptation method of the linear discriminant analysis (LDA) classifier applied to BCI systems. SVM classifier is currently the most mature application of EEG signal classification [6-7]. But it has inherent disadvantages such as the selection of kernel function is the limit by Mercer's conditions, the classification result is too sensitive for penalty factor, etc. The definition of ANN is an algorithm model of distributed parallel information processing, which imitates the behavioural characteristics of animal neural networks [8-9]. It can improve the ability of information processing by increasing the number of hidden layers, which is suitable for dealing with large data of artificial intelligence. But it is too complicated to deal with EEG data.

In this paper, we studied the basic principle of the relevance vector machine in detail [10-12]. The "one-versus-one" (OVO) mode and RVM algorithm were combined to achieve the classification of four-class motor imagery tasks. Furthermore, the RVM algorithm was modified by changing the form of kernel function. The single kernel function was replaced by the convex combination of two different kernel functions. The results of multiple experimental results show that the combined kernel function performs better than a single kernel function.

The core thought of the kernel function in RVM is to transform the concept, which is to solve the problem in low dimensional space by convert it to high dimensional space. Since it is looking for a corresponding mapping relationship, the combined kernel function may perform better because it combines the characteristics of multiple kernel functions. The generalization ability of a single kernel function is weaker than that of a combined kernel function. Therefore, two typical kernel functions, the Gaussian kernel function and the Cauchy kernel function, are selected to build the combined kernel function, and compare with both of them.

This paper is organized as follows. Firstly, we introduce the methods involved in this article, including data feature extraction, RVM classification, and will focus on the RVM part and its improvement. Secondly, the experimental results and their analysis will be presented. Finally, some conclusions and future works will be given.

## II. METHODS

### A. EEG Data Acquisition

The experimental data in this paper comes from the brain computer interface competition in 2008, and the four-class EEG data is data set2a provided by Austria Graz University. The data set consists of EEG data from 9 subjects, which is recorded in two days. For more details, please refer to the document [13]. Each session consists of four different motor imagery tasks, namely the imagination of movement of the left hand (class 1), right hand (class 2), both feet (class 3), and tongue (class 4). Each experiment is tested six times, and each experiment includes 48 motor imagery tasks. There are a total 288 trials per session. The two sessions are named E group and T group respectively. We randomly select 80% of the data for training the learning factor, the rest 20% as test data. 5 cross validation ensure the final classification result is credible.

The EEG signal recorded in the experiment by 22 AgCl electrodes. And the montage of electrode position distribution is shown in Fig.1. In the course of the experiment, the timing scheme of one trial is shown in Fig.2.

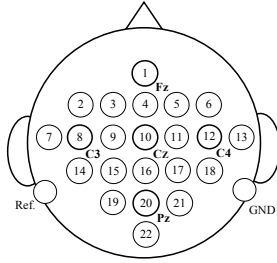


Fig.1 Electrode montage corresponding to the international 10-20 system.

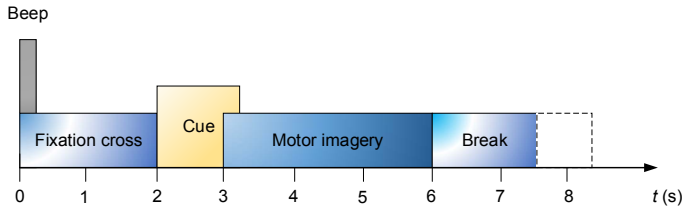


Fig.2 Timing scheme of the paradigm

### B. The process of RVM classification

The collected EEG data is processed by 3-24Hz band pass filtering. Then extract the EEG signals of the four types of imaginary tasks from the data set. And then extract the corresponding features by CSP feature extraction. Next, the processed data is divided into two groups, the training set (80%) and the test set (20%). The training set data is trained by RVM, and the training process is to find the most suitable parameters through repeated iteration and update of parameters. Finally, the model of these parameters will be sent to the test set for classification. The process of RVM classification as shown in Fig. 3.

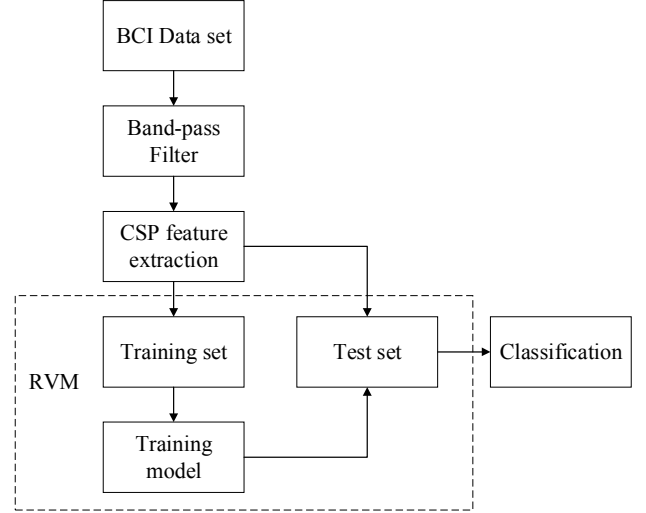


Fig. 3 The flow chart of RVM classification

### C. CSP for Feature Extraction

Common spatial pattern (CSP) is one of the most widely used feature extraction algorithms in EEG signal processing [14-18]. It computes a set of filters to find the optimal spatial projection direction, so that one-class EEG signals gain the largest projection energy from this direction, and meanwhile the other one gain the least amount of projection energy. Thus, two classes of EEG signals yield the greatest difference in the projection direction, and it is easy to separate the two-class signals.

Assuming that A and B represent two-class motor imagery tasks, the corresponding EEG signal is  $X_A$ ,  $X_B$ . Then the normalized covariance matrix of  $X_A$  and  $X_B$  can uniformly write as:

$$R_i = \frac{X_i X_i^T}{\text{trace}(X_i X_i^T)} \quad (1)$$

where  $X_i^T$  is the transpose of  $X_i$ ,  $\text{trace}(X_i X_i^T)$  is the trace of  $X_i X_i^T$ ,  $i = \{A, B\}$ .

The mean of the covariance  $R_A$  is denoted as  $\overline{R_A}$ . Similarly, we can get the corresponding matrix  $\overline{R_B}$ . Then the mixed covariance matrix  $R = \overline{R_A} + \overline{R_B}$ . The decomposition of  $R$  is given by (2)

$$R = U \Sigma U^T \quad (2)$$

where  $U$  is a matrix composed of the eigenvectors of  $R$ ,  $\Sigma$  is a matrix of eigenvalues corresponding to  $R$ .

Then construct the whitening matrix.

$$P = \Sigma^{1/2} U^T \quad (3)$$

Whitening transformation of  $\overline{R_A}$  and  $\overline{R_B}$

$$S_A = P \overline{R_A} P^T \quad S_B = P \overline{R_B} P^T \quad (4)$$

Eigenvalue decomposition of  $S_A$  and  $S_B$

$$S_A = U_A \Sigma_A U_A^T \quad S_B = U_B \Sigma_B U_B^T \quad (5)$$

It is easy to prove that  $S_A$  and  $S_B$  have the same eigenvectors, and the following equations can be obtained.

$$U_A = U_B = U \quad \Sigma_A + \Sigma_B = I \quad (6)$$

Sort the diagonal elements of  $\Sigma_A$  from large to small, and  $\Sigma_B$  changes accordingly.

Select the first  $m$  row and last  $m$  row vectors from sorted  $U$  and form a new matrix  $U_{2m}$ , thus a spatial filter  $SF_{2m}$  is designed as

$$SF_{2m} = U_{2m}^T P \quad (7)$$

For the EEG signal  $X$ , the projection matrix can be computed by the spatial filter.

$$Z = SF_{2m} X \quad (8)$$

The feature of the  $k$ -th row in projection matrix  $Z$  is calculated as

$$f_k = \log \left( \frac{\text{var}(Z_k)}{\sum_{i=1}^{2m} \text{var}(Z_i)} \right) \quad (9)$$

Where  $\text{var}()$  denotes the variance,  $k = 1, 2, \dots, 2m$ .

#### D. Relevance Vector Machine

Relevance Vector Machine (RVM) is a sparse probability model similar to SVM, which was proposed in 2000 by Michael E. Tipping [10, 12]. This is a new supervised learning method based on Bayesian model.

The model of RVM with linear output can be realized by kernel function. So the form of RVM is given as

$$y(\mathbf{x}; \mathbf{w}) = \sum_{n=1}^N w_n \phi_n(\mathbf{x}) + w_0 = \mathbf{w}^T \Phi(\mathbf{x}) \quad (10)$$

where  $\mathbf{w} = [w_0, w_1, w_2, \dots, w_N]^T$  is the weight vector,  $\mathbf{x}$  is the input sample set, and  $\Phi(\mathbf{x}) = [1, \phi_1(\mathbf{x}), \phi_2(\mathbf{x}), \dots, \phi_N(\mathbf{x})]^T$  is the kernel function matrix.

The output  $y(\mathbf{x}; \mathbf{w})$  is mapped to (0, 1) by the logistic sigmoid function, then the classification is determined.

Assuming that the samples satisfy independent and identically distributed, and  $P(\mathbf{t}|\bullet)$  is adopted the Bernoulli distribution, the likelihood function of the whole sample set can be expressed as

$$P(\mathbf{t}|\mathbf{w}) = \prod_{n=1}^N \sigma\{y(\mathbf{x}_n; \mathbf{w})\}^{t_n} [1 - \sigma\{y(\mathbf{x}_n; \mathbf{w})\}]^{1-t_n} \quad (11)$$

where  $\mathbf{w}$  is the weight vector,  $\mathbf{t} = (t_1, t_2, \dots, t_N)^T$  is the target vector, and  $t_n \in \{0, 1\}$ .

In order to improve the generalization ability of the model while preventing over-fitting, RVM defines a Gaussian prior probability distribution for each weight to achieve a smooth model.

$$p(\mathbf{w}|\boldsymbol{\alpha}) = \prod_{i=0}^N N(w_i | 0, \alpha_i^{-1}) \quad (12)$$

Where  $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \dots, \alpha_N)^T$  is the super parameter that determines the prior distribution of the weights  $\mathbf{w}$ , which

controls the degree of each weight allowing it to deviate from the zero mean value.

Calculating the posterior probability of the model parameters based on the Bayesian theorem.

$$p(\mathbf{w}, \boldsymbol{\alpha}|\mathbf{t}) = \frac{P(\mathbf{t}|\mathbf{w}, \boldsymbol{\alpha})p(\mathbf{w}, \boldsymbol{\alpha})}{p(\mathbf{t})} \quad (13)$$

If the posterior probability  $p(\mathbf{w}, \boldsymbol{\alpha}|\mathbf{t})$  is obtained, the predicted distribution of the output  $\mathbf{t}_*$  for the given test sample  $\mathbf{x}_*$  is

$$p(\mathbf{t}_*|\mathbf{t}) = \int p(\mathbf{t}_*|\mathbf{w}, \boldsymbol{\alpha})p(\mathbf{w}, \boldsymbol{\alpha}|\mathbf{t})d\mathbf{w}d\boldsymbol{\alpha} \quad (14)$$

However, the posterior distribution  $p(\mathbf{w}, \boldsymbol{\alpha}|\mathbf{t})$  cannot calculate the analytical weights. The posterior distribution can be decomposed in the form of (15).

$$p(\mathbf{w}, \boldsymbol{\alpha}|\mathbf{t}) = p(\mathbf{w}|\mathbf{t}, \boldsymbol{\alpha})p(\boldsymbol{\alpha}|\mathbf{t}) \quad (15)$$

$p(\mathbf{w}|\mathbf{t}, \boldsymbol{\alpha})$  and  $p(\boldsymbol{\alpha}|\mathbf{t})$  can be approximated using an approximation procedure based on the Laplace method:

1) Fixed the values of  $\boldsymbol{\alpha}$ , the ‘most probable’ weights  $\mathbf{w}_{MP}$  are found, i.e. the location of the mode of the posterior distribution. Since  $p(\mathbf{w}|\mathbf{t}, \boldsymbol{\alpha}) = P(\mathbf{t}|\mathbf{w})p(\mathbf{w}|\boldsymbol{\alpha})/p(\mathbf{t}|\boldsymbol{\alpha})$ , the maximum probability estimate of  $\mathbf{w}$  is equivalent to the maximum (16):

$$\begin{aligned} \ln p(\mathbf{w}|\mathbf{t}, \boldsymbol{\alpha}) &= \ln \{P(\mathbf{t}|\mathbf{w})p(\mathbf{w}|\boldsymbol{\alpha})\} - \ln p(\mathbf{t}|\boldsymbol{\alpha}) \\ &= \sum_{n=1}^N \{t_n \ln y_n + (1-t_n) \ln(1-y_n)\} - \frac{1}{2} \mathbf{w}^T \mathbf{A} \mathbf{w} + \text{const} \end{aligned} \quad (16)$$

where  $\mathbf{A} = \text{diag}(\alpha_i)$ ,  $y_n = \sigma\{y(\mathbf{x}_n; \mathbf{w})\}$ . Second-order Newton methods is applied to calculate the matrix of Hessian of (16). The ‘iteratively-reweighted least-squares’ algorithm is applied to find  $\mathbf{w}_{MP}$ .

2) Laplace’s method is the quadratic approximation of the function at the position  $\mathbf{w}_{MP}$  of the posterior probability distribution. And the Hessian matrix of (16) is

$$\nabla_{\mathbf{w}} \nabla_{\mathbf{w}} \ln p(\mathbf{w}|\mathbf{t}, \boldsymbol{\alpha})|_{\mathbf{w}_{MP}} = -(\Phi^T \mathbf{B} \Phi + \mathbf{A}) \quad (17)$$

where  $\mathbf{B} = \text{diag}(\beta_1, \beta_2, \dots, \beta_N)$  is a diagonal matrix,  $\beta_n = \sigma\{y(\mathbf{x}_n)\}[1 - \sigma\{y(\mathbf{x}_n)\}]$ . Then (17) is negated and inverted to give the covariance  $\Sigma$  for a Gaussian approximation to the posterior over weights centered at  $\mathbf{w}_{MP}$ .

At the mode of  $p(\mathbf{w}|\mathbf{t}, \boldsymbol{\alpha})$ , using (17) and the fact that  $\nabla_{\mathbf{w}} \ln p(\mathbf{w}|\mathbf{t}, \boldsymbol{\alpha})|_{\mathbf{w}_{MP}} = 0$ , we can write:

$$\Sigma = (\Phi^T \mathbf{B} \Phi + \mathbf{A})^{-1} \quad (18)$$

$$\mathbf{w}_{MP} = \Sigma \Phi^T \mathbf{B} \mathbf{t} \quad (19)$$

3) The solution of  $p(\boldsymbol{\alpha}|\mathbf{t})$  can be transformed into the maximum value of  $p(\boldsymbol{\alpha}|\mathbf{t}) \propto p(\mathbf{t}|\boldsymbol{\alpha})p(\boldsymbol{\alpha})$  for  $\boldsymbol{\alpha}$ . And the method is as follows:

$$\begin{aligned} L(\boldsymbol{\alpha}) &= \ln p(\mathbf{t}|\boldsymbol{\alpha}) = \ln \int p(\mathbf{t}|\mathbf{w})p(\mathbf{w}|\boldsymbol{\alpha})d\mathbf{w} \\ &= -\frac{1}{2} [N \ln 2\pi + \ln |\mathbf{C}| + \mathbf{t}^T \mathbf{C}^{-1} \mathbf{t}] \end{aligned} \quad (20)$$

Where  $C = B + \Phi A^{-1} \Phi^T$ .

RVM learning is ultimately reduced to maximizing the marginal likelihood function  $p(t|\alpha)$  to estimate the parameters of  $\alpha$ . In this paper, the MacKay's iterative estimation method is used to estimate  $\alpha$  [19].

$$\alpha_i^{new} = \frac{\gamma_i}{w_i^2} \quad (21)$$

$$\gamma_i \equiv 1 - \alpha_i \Sigma_{ii} \quad (22)$$

Where  $w_i$  is the mean of the  $i$ -th posterior weights calculated according to (19), and  $\Sigma_{ii}$  is the  $i$ -th diagonal element of the posterior weight covariance matrix of the current  $\alpha$  calculated according to (18). The training process of the model is to recalculate (21) and (22) and to update (18) (19) until the appropriate convergence condition is satisfied.

#### E. Improvement of kernel function

The choice of kernel function directly determines the structure of the feature space, which is the key problem of the relevance vector machine. The traditional relevance vector machine uses a single kernel function, in which the Gaussian kernel function is the most widely used, but not all cases, the Gaussian kernel function is the most suitable kernel function. In this paper, we propose a new kernel function to replace a single kernel function with convex combination of Gaussian kernel function and Cauchy kernel function. The feature data of Gaussian kernel function and Cauchy kernel function are used to map the recorded data to the feature space, and the features of different classes are maximized the difference.

The expression of Gaussian Kernel is given by (23)

$$K(X_1, X_2) = \exp\left(-\frac{\|X_1 - X_2\|^2}{2\sigma^2}\right) \quad (23)$$

The expression of Cauchy Kernel is given by (24)

$$K(X_1, X_2) = \frac{1}{\frac{\|X_1 - X_2\|^2}{\sigma} + 1} \quad (24)$$

The expression of combined kernel is given by (25)

$$K(X_1, X_2) = \lambda_1 * \exp\left(-\frac{\|X_1 - X_2\|^2}{2\sigma^2}\right) + \frac{\lambda_2}{\frac{\|X_1 - X_2\|^2}{\sigma} + 1} \quad (25)$$

The combination kernel uses the complementary properties of different kernel functions. Gaussian kernel function is good at extracting the local information of samples, so its learning ability is strong. Polynomial kernel function is good at extracting the global characteristics of samples, and has strong generalization ability. The literature [20] yield the conclusion that if  $K_1$ ,  $K_2$  are the kernel function, then the following function is the kernel function

$$K(X, Z) = K_1(X, Z) + K_2(X, Z) \quad (26)$$

$$K(X, Z) = aK_1(X, Z)$$

Therefore, we can prove the rationality of the proposed kernel function.

### III. RESULTS AND ANALYSIS

There are two sessions recorded by 9 subjects from two days. We named them as E data set and T data set respectively. Because the state of the subjects had an important influence on the accuracy of the motor imagery experiment, even if the same subject does not produce exactly the same experimental results in the same motor imagery task at different times, we analyse the two sessions separately.

TABLE I  
COMPARISON OF CLASSIFICATION RESULTS FOR T DATA SET

| Subjects | Gaussian Kernel | Cauchy Kernel | $\lambda_1$ *Gaussian-K+ $\lambda_2$ *Cauchy-K |                 |                 |
|----------|-----------------|---------------|------------------------------------------------|-----------------|-----------------|
|          |                 |               | $\lambda_1=0.7$                                | $\lambda_1=0.5$ | $\lambda_1=0.2$ |
|          |                 |               | $\lambda_2=0.3$                                | $\lambda_2=0.5$ | $\lambda_2=0.8$ |
| S1       | 66.73%          | 67.18%        | <b>72.86%</b>                                  | 71.43%          | 70.71%          |
| S2       | 46.96%          | 47.53%        | 51.07%                                         | 50.36%          | <b>51.79%</b>   |
| S3       | 75.83%          | 75.91%        | <b>78.57%</b>                                  | 78.21%          | <b>78.57%</b>   |
| S4       | 51.31%          | 51.02%        | <b>54.64%</b>                                  | <b>54.64%</b>   | 53.93%          |
| S5       | 37.68%          | 37.52%        | <b>41.79%</b>                                  | 41.43%          | 41.43%          |
| S6       | 42.14%          | 41.21%        | 43.93%                                         | 45.00%          | <b>45.71%</b>   |
| S7       | 72.92%          | 72.34%        | <b>75.71%</b>                                  | <b>75.71%</b>   | 75.36%          |
| S8       | 80.24%          | 80.84%        | 83.21%                                         | 85.00%          | <b>85.71%</b>   |
| S9       | 71.43%          | 71.84%        | 75.71%                                         | 75.71%          | <b>76.43%</b>   |
| Average  | 60.58%          | 60.60%        | 64.17%                                         | 64.17%          | <b>64.40%</b>   |

TABLE II  
COMPARISON OF CLASSIFICATION RESULTS FOR E DATA SET

| Subjects | Gaussian Kernel | Cauchy Kernel | $\lambda_1$ *Gaussian-K+ $\lambda_2$ * Cauchy-K |                 |                 |
|----------|-----------------|---------------|-------------------------------------------------|-----------------|-----------------|
|          |                 |               | $\lambda_1=0.7$                                 | $\lambda_1=0.5$ | $\lambda_1=0.2$ |
|          |                 |               | $\lambda_2=0.3$                                 | $\lambda_2=0.5$ | $\lambda_2=0.8$ |
| S1       | 64.29%          | 66.43%        | 71.07%                                          | <b>71.43%</b>   | 70.36%          |
| S2       | 42.14%          | 42.50%        | 46.07%                                          | 46.07%          | <b>46.79%</b>   |
| S3       | 78.93%          | 78.57%        | <b>82.86%</b>                                   | 82.50%          | 80.71%          |
| S4       | 59.64%          | 56.43%        | 63.21%                                          | 62.50%          | <b>63.93%</b>   |
| S5       | 46.79%          | 46.07%        | <b>52.14%</b>                                   | 51.79%          | 51.07%          |
| S6       | 41.43%          | 41.78%        | <b>47.86%</b>                                   | 46.07%          | 46.07%          |
| S7       | 74.28%          | 75.71%        | 78.21%                                          | <b>79.64%</b>   | <b>79.64%</b>   |
| S8       | 78.93%          | 80.36%        | 83.21%                                          | 83.21%          | <b>83.57%</b>   |
| S9       | 80.71%          | 82.14%        | 83.57%                                          | <b>85.00%</b>   | 83.93%          |
| Average  | 63.01%          | 63.33%        | <b>67.58%</b>                                   | <b>67.58%</b>   | 67.34%          |

Table I and Table II compare the performance of the proposed combined kernel function with the single kernel function. All computations were carried out on a Lenovo computer (CPU 3.3 GHz) and the software matlab (R2012b).

Table I is the classification results of T data set, the average accuracy rate of Gaussian kernel function and Cauchy kernel function is 60.58% and 60.60% respectively, and the highest average accuracy of the combined kernel function is 64.40%, better than the single kernel function. The highest classification results for each subject are distributed in one of the three combined kernel functions, which further proves that the combined kernel function is superior to a single kernel function.

From the experimental results of the combined kernel function, we can see that when the weight  $\lambda_1$  decreases and the weight  $\lambda_2$  increases, the classification accuracy is slightly increased. Table II also shows similar results with table I. At the same time, we can find that the average accuracy rate of the E data set is 67.58%, which is higher than that of the T data set, which also proves that the results of the same subjects in different time are not the same.

The average accuracy and standard deviation of the two sessions of each kernel function are shown in figure 4. For cleaner, we denote the Gaussian kernel, Cauchy kernel, combine kernel ( $\lambda_1=0.7$ ,  $\lambda_1=0.5$ ,  $\lambda_1=0.2$ ) with "Gauss-K", "Cauchy-K", "Cob-K1", "Cob-K2" and "Cob-K3" respectively. It can be seen that the accuracy of combined kernel function is 4%-4.19% higher than that of the single kernel function.

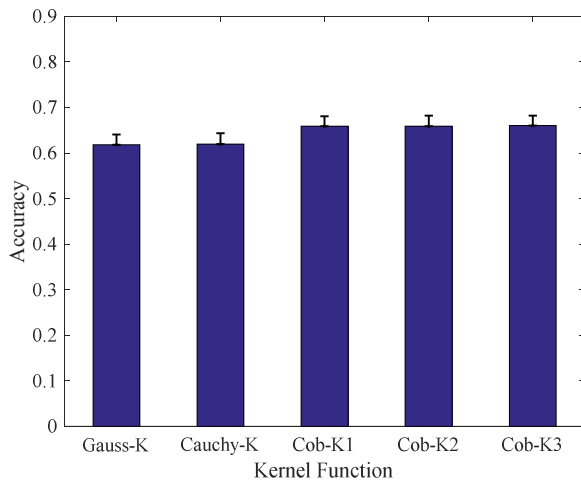


Fig.4 The average accuracy of the kernel function of two sets of data

#### IV. CONCLUSION AND FUTURE WORK

In this paper, four-class EEG signal classification is realized based on an improved RVM which replace single kernel function with combined kernel functions. The experimental results show that the RVM with improved kernel function has better classification performances, the average accuracy rate can be improved by 4% -4.19%. These results suggested that the combined kernel function is more

advantageous than the single kernel function. It also provides a new idea for the research on the pattern recognition of EEG signals.

The next step is to find a new combinatorial function based on present research. The number of combinatorial kernel functions can be added to improve the performance, rather than change the type of kernel function. The number of combined kernel functions in present work is two. It can be added to three or four, but not the more the better. It is necessary to study the optimal number of combinatorial kernel function and its generalization ability.

#### ACKNOWLEDGMENT

This work was financially supported by Natural Science Foundation of Tianjin City (15JCYBJC51800), Tianjin Key Laboratory Foundation of Complex System Control Theory and Application (TJKL-CTACS-201702) and Higher school science and technology development fund project in Tianjin (Project No. 20120829).

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